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| <b>EDUCATION</b>                            | <b>University of California</b> , San Diego, CA<br><i>Ph.D. and M.S., Computer Science &amp; Engineering</i><br>Thesis Advisor: Dr. Arun Kumar<br>Dissertation: Simplifying Data Preparation for Machine Learning on Tabular Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sept 2016 - June 2022    |
|                                             | <b>Indian Institute of Technology</b> , Indore, India<br><i>Bachelor of Technology, Computer Science &amp; Engineering</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Aug 2012 - June 2016     |
| <b>INDUSTRY<br/>EXPERIENCE</b>              | <b>IBM Corporation</b> , San Jose, CA<br><i>Staff Research Scientist</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | May 2023 - Present       |
|                                             | <i>Research Scientist</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | August 2022 - April 2023 |
|                                             | <ul style="list-style-type: none"><li>As part of Artificial Intelligence (AI) group, developing novel AI models and human-in-the-loop tools for natural language processing tasks, while reducing the barrier of entry for domain experts in adopting AI.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                          |
|                                             | <b>Microsoft</b> , Sunnyvale, CA<br><i>Research Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | June 2018 - Sept 2018    |
|                                             | <ul style="list-style-type: none"><li>Implemented computational graph-level optimizations and analytical cost model for ML operators inside Microsoft's deep learning system for inference.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                          |
| <b>ACADEMIC<br/>RESEARCH<br/>EXPERIENCE</b> | <b>Infor Corporation</b> , Berkeley, CA<br><i>Research Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | June 2017 - Sept 2017    |
|                                             | <ul style="list-style-type: none"><li>Integrated ML algorithms inside LogicBlox forecasting engine to scale training with data parallelization strategy.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                          |
|                                             | <b>University of California, San Diego</b><br><i>Graduate Student Researcher</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Sept 2016 - June 2022    |
|                                             | <ul style="list-style-type: none"><li><b>CategDups</b>. Presents novel data artifacts, benchmarks, and empirical analyses to help ML practitioners prioritise their effort in cleaning Categorical duplicates and Automated machine learning(AutoML) developers to build better deduplication workflows.</li><li><b>SortingHat</b>. Created the first benchmark for ML feature type inference by leveraging database schema semantics to objective quantify and substantially improve the accuracy of the task. This helps to objectively validate and improve AutoML platforms.</li><li><b>ML Data Prep Zoo</b>. Vision of how we leverage ML for systematically standardizing and automating data preparation for ML with a zoo of labeled dataset and ML models.</li><li><b>SpeakQL</b>. Developed a system for making spoken SQL querying effective and efficient. The speech-driven interface allows the users to query in any domain with infinite vocabulary using interactive query correction.</li><li><b>Hamlet</b>. Analyzed the accuracy effects of joins on high-capacity ML algorithms, when learning over normalized data to reduce the data sourcing burden for ML.</li></ul> |                          |
|                                             | <b>University of Alberta, Canada</b><br><i>Research Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | May 2015 - July 2015     |
|                                             | <ul style="list-style-type: none"><li>Developed a Big Data Adaptor as Eclipse plugin which efficiently handles GitHub's data dump. The tool saves developers' time and effort in data prep for GitHub analysis.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                          |

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| <b>PUBLICATIONS SUMMARY</b>       | <p>Accepted Papers at top-tier conferences and workshops: 13</p> <p>US Patents: 1 approved and 1 submitted</p> <p>Manuscripts, Whitepapers, and Technical Reports: 5</p> <p>Number of citations: 314 and h-index: 10 (as per <a href="#">Google Scholar</a> in Aug 2026)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>PEER-REVIEWED PUBLICATIONS</b> | <p><i>SemEval-2026 Task 8: MTRAGEval: Evaluating Multi-turn RAG Conversations</i><br/> Sara Rosenthal, <b>Vraj Shah</b>, Yannis Katsis, and Marina Danilevsky.<br/> Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026), Association for Computational Linguistic ACL 2026   <a href="#">Paper</a></p> <p><i>MTRAG-UN: A Benchmark for Open Challenges in Multi-Turn RAG Conversations</i><br/> Sara Rosenthal, Yannis Katsis, <b>Vraj Shah</b>, Lihong He, Lucian Popa, Marina Danilevsky<br/> Association for Computational Linguistics ACL Findings 2026   <a href="#">Paper</a></p> <p><i>Activated LoRA: Fine-tuned llms for intrinsics.</i><br/> Kristjan Greenewald, Luis Lastras, Thomas Parnell, <b>Vraj Shah</b>, Lucian Popa, Giulio Zizzo, Chulaka Gunasekara, Ambrish Rawat, David Cox<br/> Annual Conference on Neural Information Processing Systems. NeurIPS 2025.   <a href="#">Paper</a></p> <p><i>MTRAG: A Multi-Turn Conversational Benchmark for Evaluating Retrieval-Augmented Generation Systems</i><br/> Yannis Katsis, Sara Rosenthal, Kshitij Fadnis, Chulaka Gunasekara, Young-Suk Lee, Lucian Popa, <b>Vraj Shah</b>, Huaiyu Zhu, Danish Contractor, Marina Danilevsky<br/> Transactions of the Association for Computational Linguistics. TACL 2025   <a href="#">Paper</a></p> <p><i>How do Categorical Duplicates Affect ML? A New Benchmark and Empirical Analyses.</i><br/> <b>Vraj Shah</b>, Thomas Parashos, Arun Kumar.<br/> Proceedings of the VLDB Endowment. VLDB 2024   <a href="#">Paper</a></p> <p><i>Towards Benchmarking Feature Type Inference for AutoML Platforms.</i><br/> <b>Vraj Shah</b>, Jonathan Lacanlale, Premanand Kumar, Kevin Yang, Arun Kumar.<br/> Proceedings of the 2021 International Conference on Management of Data. SIGMOD 2021   <a href="#">Paper</a></p> <p><i>SpeakQL: Towards Speech-driven Multimodal Querying of Structured Data.</i><br/> <b>Vraj Shah</b>, Side Li, Arun Kumar, Lawrence Saul.<br/> Proceedings of the 2020 International Conference on Management of Data. SIGMOD 2020   <a href="#">Paper</a></p> <p><i>Demonstration of SpeakQL: Speech-driven Multimodal Querying of Structured Data.</i><br/> <b>Vraj Shah</b>, Side Li, Kevin Yang, Arun Kumar, Lawrence Saul.<br/> Proceedings of the 2019 International Conference on Management of Data. SIGMOD 2019 (<i>Demo track</i>)   <a href="#">Paper</a></p> <p><i>The ML Data Prep Zoo: Towards Semi-Automatic Data Preparation for ML.</i><br/> <b>Vraj Shah</b>, Arun Kumar.<br/> Proceedings of the 3rd International Workshop on Data Management for End-to-End Machine Learning. DEEM Workshop SIGMOD 2019   <a href="#">Paper</a></p> <p><i>SpeakQL: Towards Speech-driven Multi-modal Querying.</i><br/> <b>Vraj Shah</b>.<br/> Proceedings of the 2019 International Conference on Management of Data. SIGMOD 2019 <b>Awarded Second Runner-up</b>   <a href="#">Paper</a></p> <p><i>Are Key-Foreign Key Joins Safe to Avoid when Learning High Capacity Classifiers?</i><br/> <b>Vraj Shah</b>, Arun Kumar, Xiaojin Zhu.<br/> Proceedings of the VLDB Endowment. VLDB 2018   <a href="#">Paper</a></p> |

*SpeakQL: Towards Speech-driven Multi-modal Querying.*

Dharmil Chandarana, **Vraj Shah**, Arun Kumar, Lawrence Saul.

Proceedings of the 2nd Workshop on Human-In-the-Loop Data Analytics. HILDA Workshop SIGMOD 2017 | [Paper](#)

*GitHub's Big Data Adaptor: An Eclipse Plugin.*

Ali Sajedi, **Vraj Shah**, Eleni Stroulia.

Proceedings of the 25th Annual International Conference on Computer Science and Software Engineering. CASCON 2015 | [Paper](#)

## PATENTS AND RESEARCH ARTIFICATS

Human-AI Collaborative Error Pattern Discovery in LLM Responses  
US Patent (Submitted 2026).

Muntasir Wahed, Marina Danilevsky Hailpern, Ioannis Katsis, **Vraj Paragbhai Shah**,  
Huaiyu Zhu

Speech Based Structured Querying.

US Patent Approved 2021.

Arun Kumar, **Vraj Shah**, Dharmil Chandarana | [Patent #11086937](#)

A Zoo of Labeled Datasets and Pre-trained AI models for ML data prep tasks.

**Vraj Shah**, Arun Kumar | [Github](#)

## MANUSCRIPTS AND TECHNICAL REPORTS

*A Library of LLM Intrinsic for Retrieval-Augmented Generation*

Marina Danilevsky, Kristjan Greenewald, Chulaka Gunasekara, Maeda Hanafi, Lihong  
He, Yannis Katsis, Krishnateja Killamsetty, Yatin Nandwani, Lucian Popa, Dinesh  
Raghu, Frederick Reiss, **Vraj Shah**, Khoi-Nguyen Tran, Huaiyu Zhu, Luis Lastras  
2025 | [Technical Whitepaper](#)

*Simplifying Data Preparation for Machine Learning on Tabular Data.*

**Vraj Shah** | UCSD CS Ph.D. Dissertation 2022. | [Manuscript](#)

*Improving Feature Type Inference Accuracy of TFDV with SortingHat*

**Vraj Shah**, Kevin Yang, Arun Kumar | 2020 | [Technical Whitepaper](#)

**Research Impact.** Models explored for production use by TensorFlow Data Validation  
in collaboration with Google.

*A Label Efficient Strategy for Source-Free Specialized Domain Adaptation*

**Vraj Shah**, Yannis Katsis, Lucian Popa, Salim Roukos, Tejaswini Pedapati. | Under  
Submission

*Stop That Join! Discarding Dimension Tables when Learning High Capacity Classifiers*

**Vraj Shah**, Arun Kumar, Xiaojin Zhu | [Technical Report](#)

## PEER-REVIEW EXPERIENCE

Reviewed a total of 80 papers as a *Program Committe* member for top-tier conferences  
and workshops.

- The International Journal on Very Large Data Bases (VLDB Journal): 2025
- International Conference on Management of Data (ACM SIGMOD): 2025, 2024
- First Workshop on Data-AI Systems (DAIS) at IEEE International Conference  
on Data Engineering (ICDE): 2025
- International Conference on Very Large Databases (VLDB): 2024, 2022
- International Conference on Data Science & Management of Data (ACM CODS-  
COMAD): 2024
- Table Representation Learning Workshop (TRL) at Conference on Neural Infor-  
mation Processing Systems (NeurIPS): 2022

- Workshop on Data Management for End-to-End Machine Learning (DEEM) at ACM SIGMOD: 2022

*External Reviewer:*

- VLDB: 2019
- VLDB (Demo Track): 2018

## AWARDS

- Awarded *Second Runner-up* at ACM SIGMOD Student Research Competition in 2019, out of ~80 students from across the world.
- Received NSF Travel Award to attend ACM SIGMOD 2019.
- Received NSF Travel Award to attend VLDB 2018.
- Received Microsoft Travel Award to attend ACM SIGMOD 2017.
- Awarded with Research Experience program honor for poster presentation at *University of Alberta International Research Symposium* in 2015.
- Awarded Canadian MITACS Globalink Award for Research Internship in 2015.
- Awarded with DAAD (German Academic Exchange Service) Working Internships in Science and Engineering (WISE) Fellowship in 2015.
- Attained an All India Rank of 3391 in the highly competitive IIT-JEE (common entrance examination for top Indian engineering schools) in 2012, out of 4,79,651 students.

## GRANT SUPPORT

My research projects during my Ph.D. have received funding and support from several grants and gifts. Co-wrote the following grants and successfully delivered on them: (1) NSF award IIS-1816701 (2) NSF Convergence Accelerator under award OIA-2040727 (3) Faculty Research Award from Google Research (4) Amazon Research Award.

## TALKS

- SIGMOD 2021. Towards Benchmarking Feature Type Inference for AutoML Platforms.
- SIGMOD 2020. SpeakQL: Towards Speech-driven Multimodal Querying of Structured Data.
- SIGMOD 2019. Demonstration of SpeakQL System for Speech-driven Multimodal Querying of Structured Data.
- DEEM Workshop at SIGMOD 2019. The ML Data Prep Zoo: Towards Semi-Automatic Data Preparation for ML.
- VLDB 2018. Are Key-Foreign Key Joins Safe to Avoid when Learning High Capacity Classifiers?
- HILDA Workshop at SIGMOD 2017. Towards Speech-driven Multi-modal Querying.

## TEACHING EXPERIENCE

- Teaching Assistant - [DSC 102: Advanced Data Analytics Systems](#) Winter 2020  
Co-created the first edition of the course programming assignments (PAs), which includes data exploration with AWS and Dask, and feature engineering and model selection with Spark. The PAs have been used by 450+ UCSD students so far and are now used in every DSC102 course offering.
- Teaching Assistant - [CSE 132C: Database System Implementation](#) Spring 2020